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Bio-inspired minimal hardware implementation of spike-time dependent plasticity for intelligent spiking neural networks

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E-mail: marcelo.rozenberg@cnrs.fr**Keywords:** neuromorphic hardware, spiking neural networks, spike-timing dependent plasticity

Abstract

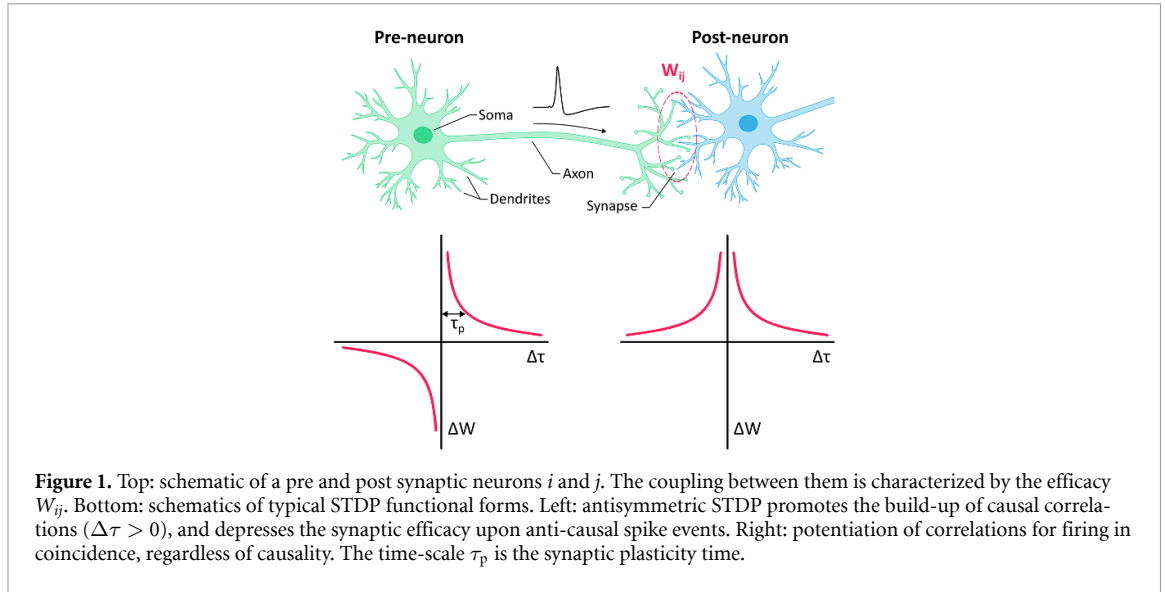
We introduce a hardware circuit model that implements spike-time dependent plasticity (STDP) to endow spiking neural networks with learning capabilities. Our circuit model is characterized as both minimal and bio-inspired, due to its simplicity and to a novel active dendrite compartment that mimics the synaptic potentiation mechanism. The active dendrite consists of an integrate-and-fire stage which produces a train of pulses whose number is inversely related to the timing between pre- and post-synaptic spikes. The dendrite pulses modulate in a reliable manner the synaptic efficacy (conductance) that we implemented with a *digipot*, considered as an idealized non-volatile memristor. We demonstrate the behavior of the circuit by implementing a minimal spiking neuron model of associative learning by STDP, which is analog to the classic conditioning experiment of Pavlov's dog.

1. Introduction

Understanding learning is one of the greatest open challenges in science and is key to endow intelligence to systems and devices. Contributions to this problem are multidisciplinary, originating in fields that range from neuroscience to computer science, psychology, and more. The mechanistic origin of learning, at the level of neuronal circuits, is based on the widely accepted dogma that it is rooted in the modification of synaptic efficacies due to experience. The evolution of the synaptic strengths between neurons tailors the architecture of neural networks and enables animals and humans to acquire new motor and cognitive functions. While this topic is a matter of intense research, a main paradigm for the modification of synaptic couplings is the experimentally observed spike-timing dependent plasticity (STDP) [1–3]. This mechanism is often viewed as Hebb's famous statement 'cells that fire together wire together' [4].

STDP can be considered as the natural mechanism for a pair of neurons to build-up correlations between their spiking activities. In the celebrated Pavlov's dog experiment [5], an initially neutral stimulus, such as the ringing of a bell, is presented along an unconditioned one, such as food. After a few 'teaching' sessions where the two stimuli are presented together, the dog learns that a bell is associated with food, and starts drooling at the sole ringing. Hence a correlation has formed between two previously unrelated phenomena: the neural response to the sound of a bell and the neural stimulation of a saliva producing gland.

That celebrated experiment is an example of supervised learning, as it requires the action of a 'teacher'. Nevertheless, the STDP mechanism is also associated to a more challenging type of learning, called unsupervised. In this case, an animal, or a neural network, learns from its own experience, that is, by trial and error. In that case the neural correlations are built as consequence of a self-produced mechanism such as pleasure or dislike. This should not be confused with an innate behavior, which may be



due to a pre-wiring of the network. Typical examples of unsupervised behavior are classification tasks. For instance, discovering tasty fruits or clustering in the classroom, such as when children go about distinguishing friends from foes in pre-school.

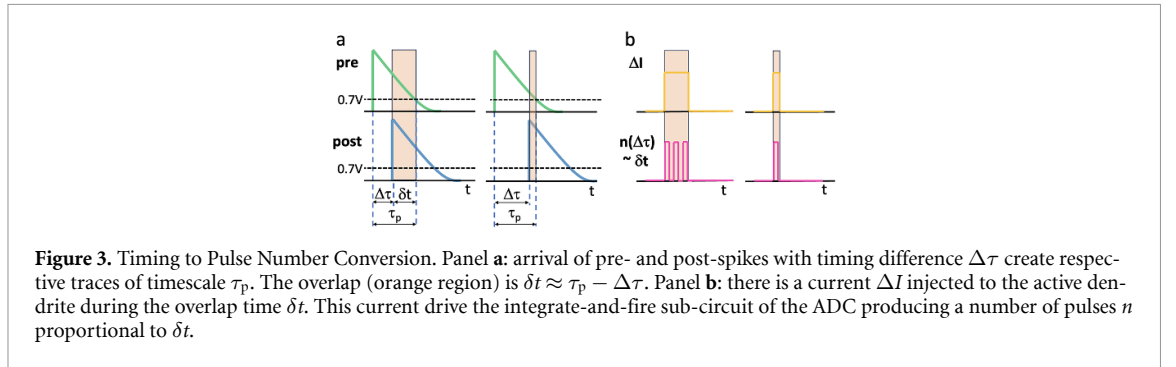
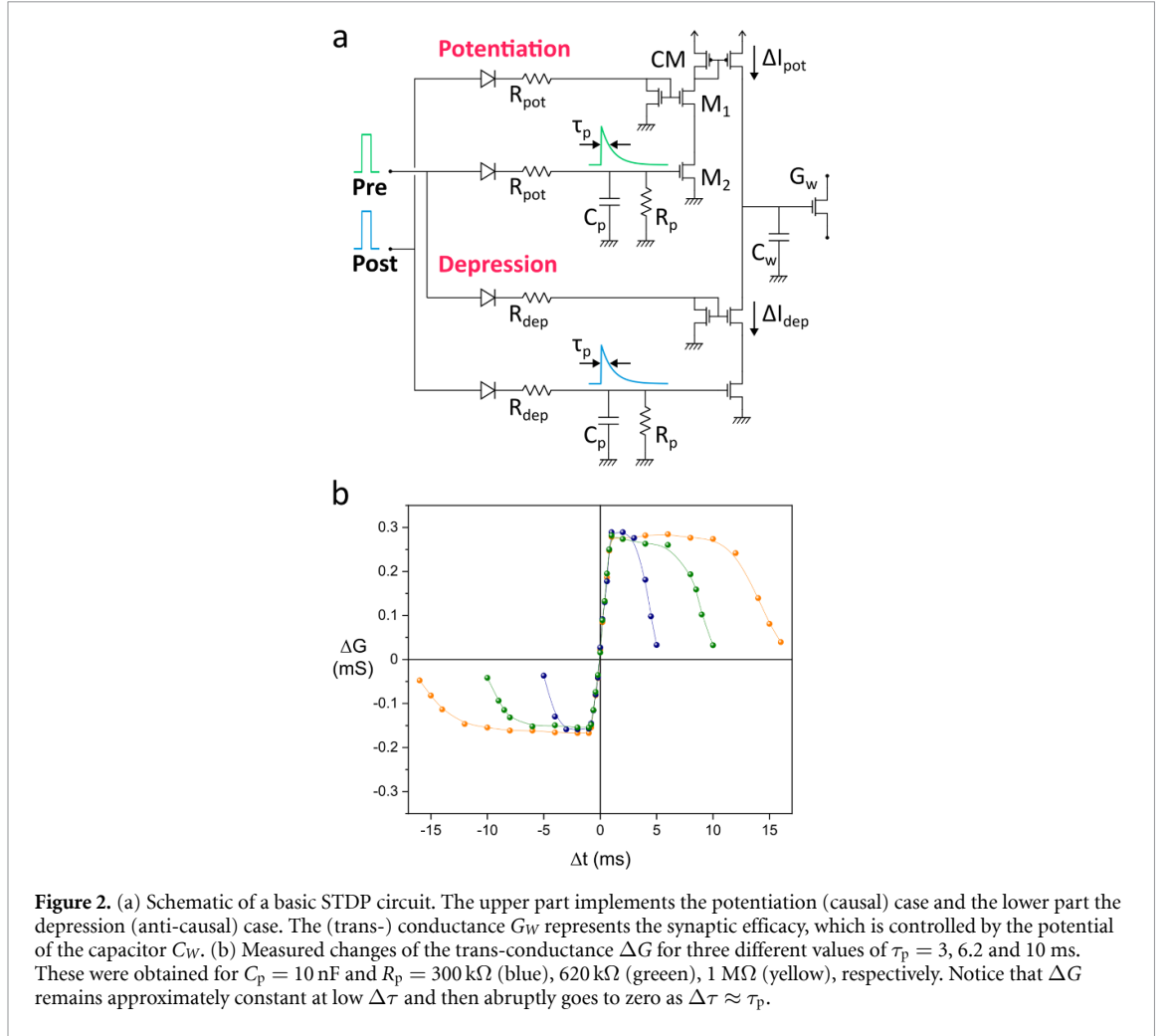
The STDP mechanism can be mathematically formalized by defining the change in the synaptic efficacy ΔW as a function to the relative timing $\Delta\tau$ between the spike emission events of the two neurons sharing that synapse. Concretely, let $\Delta\tau_{ij}$ denote the time difference $t_j - t_i$ between spikes fired by the pre-synaptic neuron i and the post-synaptic neuron j , coupled by the synaptic efficacy W_{ij} (see figure 1). The type of STDP is characterized by the function $\Delta W_{ij}(\Delta\tau_{ij})$. Typically, this function is peaked at small values of the argument, as it enhances correlations between the firing events of two neurons. If the change ΔW is positive, it is called potentiation, as the coupling is made stronger. In contrast, if it is negative, it is called depression, as the influence of one neuron on the other is decreased. For instance, if the pre-neuron fires shortly before the post-neuron and ΔW is large and positive, then the STDP mechanism contributes to build-up a positive correlation between the firing of i and j , i.e. enhances a causal relation between them. Figure 1 shows two common instances of STDP, and this can be generalized in multiple ways [6].

The biological underpinning of the STDP mechanism is still a matter of study [7]. For the potentiation of a synapse, one may argue that as a pre-synaptic action potential arrives to the synapse, it produces a temporary sensitization of the junction during a plasticity time-scale τ_p . Then, if a post-synaptic action potential occurs within the τ_p window, it may produce a spike signal that propagates *upstream* along the post-synaptic dendrite. This back propagating potential may eventually hit the still sensitized synaptic junction and induce an increase in its efficacy. Motivated by such a process, in this work we shall propose a phenomenological bio-inspired model for STDP that we implement in a minimal hardware circuit. To that end we shall adopt a recently introduced minimal hardware model of an excitable (i.e. spiking) system [8, 9].

2. Model

2.1. Basic STDP circuit

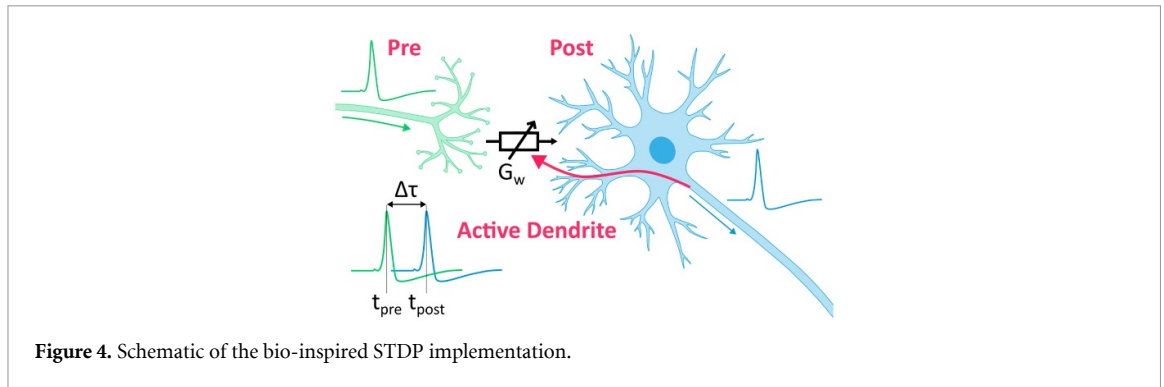
We shall first describe a rather schematic circuit implementation of STDP, which has some common features found in many related previous work [10–15]. This basic circuit will serve us as a reference to later introduce our novel bio-inspired hardware model. In a typical implementation of STDP, as shown in the circuit of figure 2, and the schematic diagram in panel a of figure 3, the arrival of a pre-synaptic spike V_{pre} creates a voltage trace V_p that decays within a characteristic plasticity time $\tau_p \approx R_p C_p$. This trace enables the transistor M2 during τ_p , so long $V_p \gtrsim 0.7V$, the threshold for polarization of M2. If a post-synaptic spike V_{post} arrives, it will generate a similar trace that enables the second transistor M1 during a τ_p . If the time difference $\Delta\tau$ between these two arrivals is smaller than τ_p , then both M1 and M2 will be simultaneously enabled during a certain overlap time $\delta t \sim (\tau_p - \Delta\tau) > 0$. During that time window δt , a current $\Delta I_{pot} \approx (V_{post} - 0.7V)/R_{pot}$ will flow out of the current mirror CM. Hence, an amount of



charge $\delta q \approx \Delta I_{\text{pot}} \delta t$ will be added to the weight capacitor C_W , increasing the (trans-)conductance G_W in a certain ΔG which corresponds to the ΔW of figure 4. The conductance G_W represent the synaptic coupling and its increase provides a *potentiation* of its efficacy. Note that the amount ΔG is controlled by the magnitude of the resistors R_{pot} (and R_{dep} for depression), as shown in figure 2.

A similar analysis can be done for the synaptic *depression* case that corresponds to the anti-causal arrival of spikes. This is, when the pre-synaptic spike arrives *after* the post-synaptic one. The sole circuit difference is the configuration of the current mirror that produces a negative ΔI . Therefore there is a discharge of C_W and a decrease of the synaptic conductance G_W by a certain $-\Delta G$, whose value can be controlled with R_{dep} .

The circuit just described was implemented and measured, and the behavior for three different values of τ_p is shown in figure 2. We observe that, consistent with the previous discussion, $\Delta G(\Delta\tau)$ shows a dependence on the time-scale τ_p . However, one may consider a shortcoming of this simple STDP implementation that ΔG does not show a significant dependence on $\Delta\tau$, as it exhibits a step-like behavior.

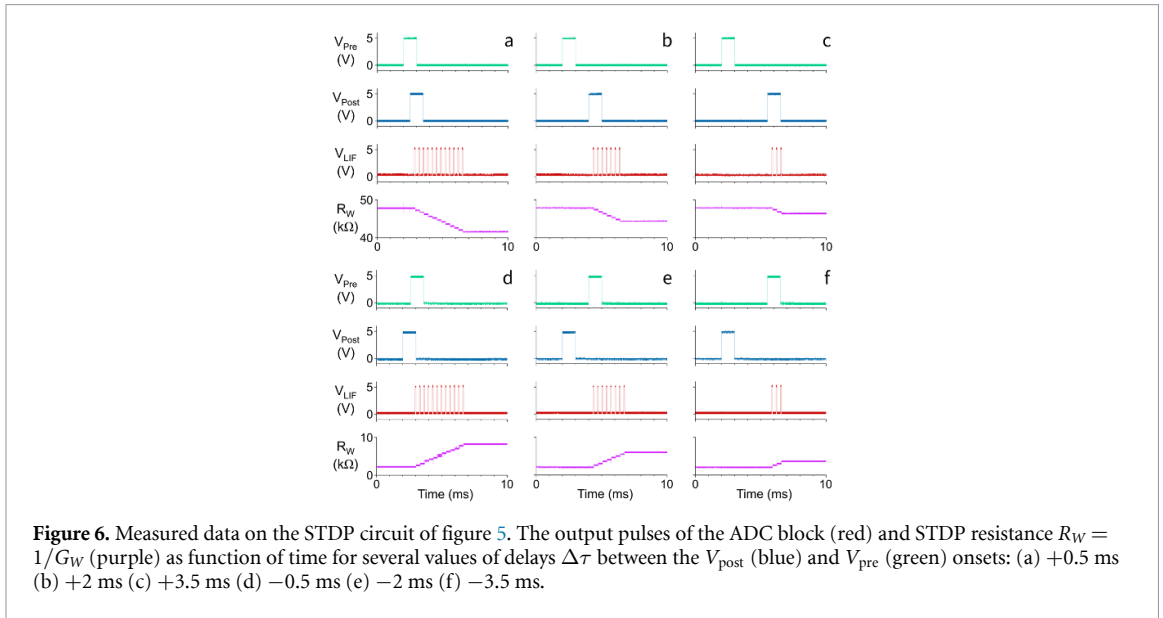
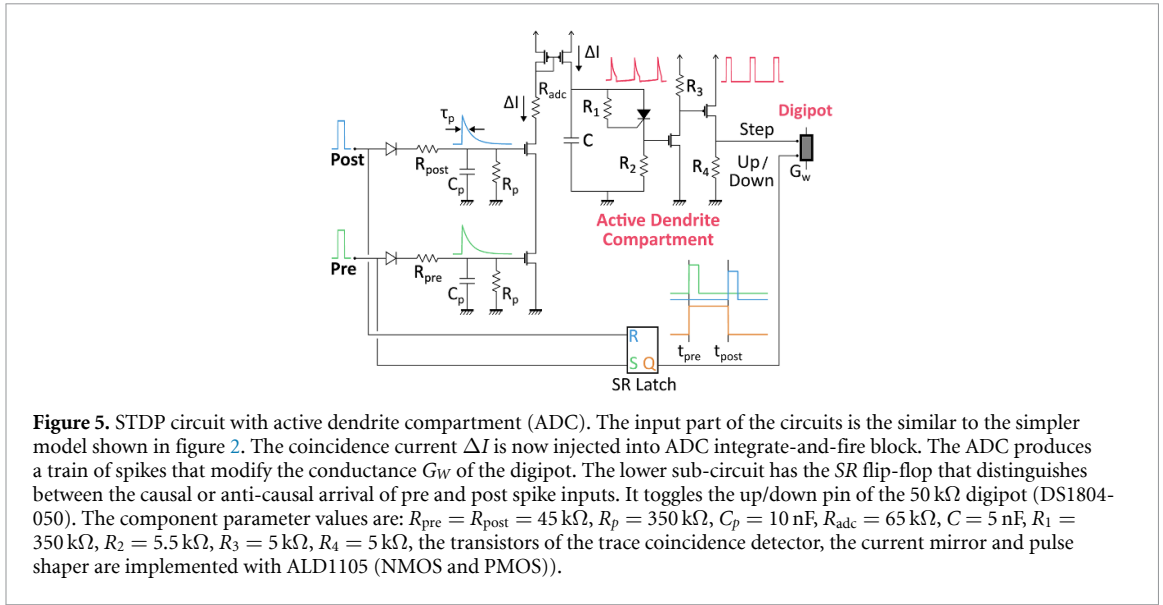


This is unlike typical theoretical models of STDP which show a decaying behavior (cf figure 1). Another shortcoming, perhaps more important, is rather practical as it is difficult to maintain the charge of C_W beyond the order of the second, due to its intrinsic leak. As we shall see below, these shortcomings will be significantly improved by our bio-inspired STDP model circuit.

2.2. Bio-inspired circuit with active dendrite compartment (ADC)

Our new implementation is schematically shown in figure 4. It introduces two main features with respect to the basic STDP model. Firstly, we replace the capacitor-plus-transistor that model the tunable synaptic conductance. Such a combination is hard to control due to both, the threshold behavior of the transistor and the leak of C_W . As we mentioned, the later is specially problematic, as it quickly erases the eventual learning (although it may find applications in short term memory). Our practical solution is to replace the capacitor plus transistor by a digital potentiometer (digipot). The digipot resistance (and conductance) can be easily changed, up- or downwards, between 0 to $R_{\max}$ in a number N of equal discrete steps δr , such that $R_{\max} = N\delta r$. It is interesting to mention that a digipot can also be considered an idealized *non-volatile* memristor, i.e. a resistive component whose value can be tuned by application of pulses [16]. Memristive materials and devices are under intense current research for energy-efficient artificial intelligence hardware [17–19]. Our digipot solution has the merits of wide availability, reliability and simple implementation. It therefore serves our present goal of introducing a novel STDP synaptic circuit model. Nevertheless, the digipot component may eventually be replaced by non-volatile memristors, if low-energy and small footprint are required. The second and most important feature that we introduce in our model is an active dendrite compartment (ADC, not to be confused with analog-digital converter). This is a bio-inspired mechanism that allows to modify the synaptic weight depending on the timing difference $\Delta\tau$. The ADC is modeled as an excitable sub-circuit that sends an *upstream* spiking signal. To implement these dendritic pulses we adopt an *integrate-and-fire* circuit model based on a *volatile* memristor device made of a thyristor-plus-resistor, which was recently introduced [8, 9]. The ADC translates the delay $\Delta\tau$ into a number of spikes that are used to modify the conductance value of the synaptic digipot. As we shall see, an important feature of our circuit model is that, consistent with theoretical STDP models and its experimental observation, the smaller the $\Delta\tau$, the stronger the ‘upstream’ signal and the larger the conductance change of the synapse. This bio-mimetic feature is rarely implemented in hardware models. Our bio-inspired STDP circuit model is shown in figure 5. Similarly to the basic circuit that we described before, this circuit also has two RC pairs at the input to provide the two traces $V_{\text{pre}}^p(t)$ and $V_{\text{post}}^p(t)$ of the arriving spikes. As before, the traces start at the arrival time of the spikes, t_{pre} and t_{post} , and both decay exponentially with the time-constant $\tau_p \approx R_p C_p$ as shown in panel a of figure 3.

The pre- and post- traces, as in the previous circuit, enable the gates of two respective transistors that are in series. When the traces overlap and they are simultaneously above 0.7V, the input side of the current mirror is enabled. In this case, there is a current $\Delta I \approx V_+/R_{\text{adc}}$ that flows during the overlap time δt , as shown in panel b of figure 3. The current mirror reproduces the current ΔI at its output. Hence, the ADC compartment gets as input the same current ΔI during a time δt . The ADC is modeled as an excitable integrate-and-fire unit, similar as that introduced in [8]. Thus, the input ΔI to the ADC produces a train of equally spaced spikes during the period δt , whose frequency is controlled by the intensity of ΔI (i.e. integrate-and-fire model). Hence, the number n of spikes in the train is $n \propto \delta t$. Those spikes are subsequently reshaped into 0–5 V pulses by the two transistors as shown in figure 5 [8, 20], for proper input to the digipot. An example of this is displayed in figure 6: the first and second

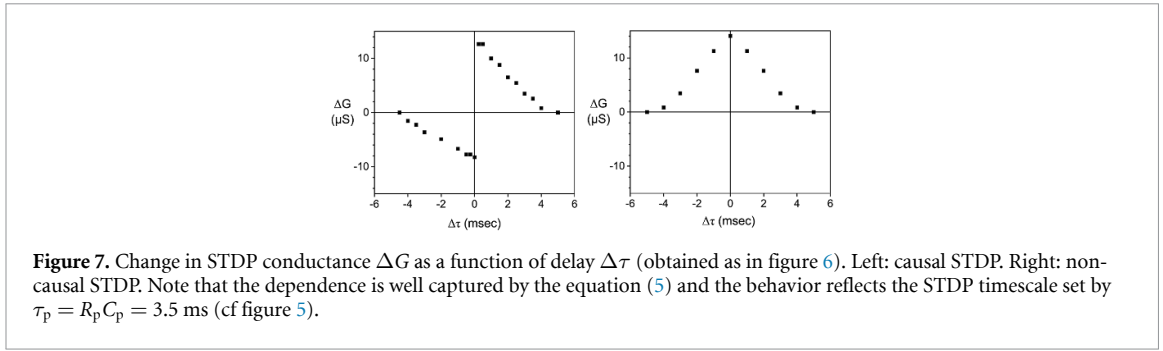


row from the top show the pre- and post-synaptic incoming spikes (in green and blue, not to be confused with their respective traces), and the third row shows the resulting spike-pulse trains (in red) at the output of the ADC. Crucially, we observe that the number of pulses is inversely related to the time-difference $\Delta\tau$ between the arrival of the pre- and post-synaptic spike, consistent with our previous discussion. When $\Delta\tau \gtrsim \tau_p$ there are no more output spikes, where τ_p is the plasticity timescale. Since, as noted before, $\delta t \approx \tau_p - \Delta\tau$ the approximate behavior of the number of pulses in the train is

$$n(\Delta\tau) = n_0 [1 - |\Delta\tau| / (k \tau_p)] \quad (1)$$

where n_0 is a constant (that depends on ΔI and model parameters) and k is of order one.

Lets now estimate the expected change in the STDP conductance ΔG_W that this implementation should provide. The digipot is characterized by its constant resistive change step δr . Hence, for a digipot at an arbitrary initial resistance state R_W , an incoming train of n pulses changes R_W in a simple linear fashion, i.e. $\Delta R_W = \mp n \delta r$ (provided the final resistance $R_W + \Delta R_W$ does not reach any of the digipot limits). The negative sign is for causal arrival, i.e. the pre-spike arrives before the post-spike, and the plus of the anti-causal. This behavior is shown in the measured data of R_W (in purple) in figure 6, for both cases, causal and anti-causal.



We now need to estimate the corresponding change ΔG_W of the STDP conductance when it is in an initial state $G_W = 1/R_W$. We have,

$$\Delta G_W = \frac{1}{R_W \mp n\delta r} - \frac{1}{R_W} \quad (2)$$

$$\approx G_W \left(1 \pm n \frac{\delta r}{R_W} \right) - G_W. \quad (3)$$

Hence,

$$\Delta G_W \approx \pm n \frac{\delta r}{R_W^2} = \pm n \delta r G_W^2 \quad (4)$$

therefore, the change in the STDP conductance is also approximately linear in n , so long the initial value of R_W is not close to the low resistance end. In other words, it is much larger than the resistance step, i.e. the initial $R_W \gg \delta r$. This condition can always be fulfilled by adding a small fixed series resistor to R_W , bigger than the resistive step.

We should also note, from equation (4), that the change ΔG_W has a non-linear dependence in the initial value of $G_W = 1/R_W$. Specifically, given a number n of pulses in the train, the corresponding ΔG_W will be relatively larger, the larger the initial value of G_W . Within a learning scheme this qualitatively means that the initial learning stages (from low values of G_W) are relatively ‘harder’ than the later ones.

Using equation (1), we can finally estimate the change $\Delta G_W(\Delta\tau)$. We have,

$$\Delta G_W(\Delta\tau) \propto \pm n \sim \pm n_0 [1 - |\Delta\tau| / (k \tau_p)] \quad (5)$$

which is a linearly decreasing function of $\Delta\tau$, in good agreement with the measured data shown in figure 7, and qualitatively similar to the behavior depicted in figure 1.

We need to discuss one more feature of the model circuit of figure 5, namely the RS flip-flop that allows to keep the information of the order of arrival of the pre- and post-synaptic spikes. This sub-circuit is optional. If it is not included, the STDP model provides a realization of the, ‘non-causal’, second type of function $\Delta W(\Delta\tau)$, as shown in figures 1 and 7. In this case, the arrival order, i.e. causality, is not important. Thus, the $(\pm)$ drops out from the expressions above and $\Delta G(\Delta\tau) > 0$ always increases.

On the other hand, if causality is deemed important, the sub-circuit allows to toggle the digipot between its up/down mode. We implement this with a SR flip-flop that works as follows: assume the initial inputs and output are all 0 and first consider a causal sequence. A logic 1 ($V_{pre} = 5V$) arrives at S (set), then the output Q changes to 1. A successive logic 1 arrives at R (reset), then the output Q changes to 0. Hence, the digipot receives a logic 0 and goes to the down mode, ready to decrease its resistance (cf fourth row of top panels in figure 6) when the spike train of the active dendrite arrives (cf third row panels in figure 6). The output $Q = 0$ is maintained so long the R and S do not receive spikes. In the anti-causal case, i.e. pre arrives after post, a 1 is received first at R and then at S. The output Q of the flip-flop goes to 1 and it remains, so long no further pulses arrive. This sets the digipot to the up mode. The increase of resistance for the anti-causal case can be observed in the fourth row of the bottom panels of figure 6. We implemented the SR flip-flop using two NOR gates.

One other feature that we should also mention about the circuit relates to an important bio-inspired feature, the ‘dopamine effect’, which is key for reinforcement learning. Dopamine is a neurotransmitter that modulates the intensity of the synaptic changes. This is captured by our circuit in the following manner: The intensity of the current ΔI controls the frequency of the pulses generated by the ADC.

Hence, given a time-delay between the pre- and post- pulses, the magnitude of ΔI controls how many pulses will be produced, ie the size of the STDP synaptic conductance change. The key point is that the intensity of the current is controlled by the resistance R_{adc} , as was described before. Therefore, the magnitude of the conductance $1/R_{\text{adc}}$ can be qualitatively assimilated to the effect of dopamine concentration, as it modulates the strength of the synaptic change. This feature may be exploited in implementations of networks with reinforcement learning by replacing R_{adc} by a tunable resistance (such as a second digipot or a non-volatile memristor).

3. Measured circuit behavior

We now illustrate the operation of the STDP with two different paradigmatic examples.

3.1. Synaptic strength modulation

In the first case we input two spike trains pre- and post- with a slightly different frequency. This produces a beating effect, namely the pre-spike precedes the post-spike during some time (causal $\Delta\tau > 0$) and then the situation alternates and the post-spike precedes the pre-spike (anti-causal $\Delta\tau < 0$). Hence, the change in the synaptic resistance (or conductance) periodically increases and decreases. This can be seen in figure 8, which demonstrates the phenomenon and allows to appreciate the stability of the circuit behavior.

3.2. Classical conditioning

We now demonstrate a basic application of the STDP behavior in associative learning, as in the classic example of Pavlov's dog [5]. In figure 9 we show the detailed circuit implementation, where we can see the spike (pulse) generators in red and green for the 'meat' and 'bell' for the unconditioned and conditioned stimuli, respectively. The first one has a large intrinsic synaptic efficacy that directly drives the 'dog's salivation' neuron response (blue) with a strong synaptic current (orange). In contrast, the dog's response to the bell stimulus is controlled by the STDP conductance G_W , which is weak before the learning. Hence, the corresponding synaptic current (purple) is also small and cannot drive the salivation. Upon activation of the stimuli pulses, the STDP mechanism will associate the provoked dog's behavior (blue) to the initially innocuous stimulus (green) that precedes it. The correlated spiking activity of salivation and bell creates an increase of the synaptic conductance G_W , following the STDP mechanism described above. When G_W efficacy increases enough, the bell stimulus eventually become able to produce a salivation response. We say that the dog has learned to associate the sound of the bell to the food, and now the bell sound drives the dog's drooling. By the same mechanism, the anti-correlated stimulation of bell and salivation would lead to the 'forgetting' of the previously learned behavior. The measured data that demonstrate this phenomenon is shown in figure 10, where we observe three phases: in the first phase, only the innocuous stimulus (bell) is active, shown in green spikes, and fails to produce a response (salivation) in blue. Then, in the 'learning' phase, both stimuli, innocuous and unconditioned (meat) are active. The former produces the response (salivation), due to the intrinsic strong synaptic efficacy (orange arrow). The key feature for learning is that the response (salivation) is causally associated to the innocuous stimulation (bell), i.e. the green spikes shortly precede the blue ones. This causal correlation produces the increase of the synaptic efficacy (purple arrow) between the green and blue neurons. As a result, after the associative learning phase, we obtain the 'conditioned' behavior, where the 'dog' has learned to respond to the previously innocuous bell stimulus, i.e. the green-neuron spikes are now provoke themselves the response of the blue neuron.

3.3. Energy requirements

The quantitative comparison of the energy consumption of our STDP circuit to other analog proposals is unfortunately not possible due to the lack a consistent benchmarks and lack of reports. Our STDP circuit of figure 5 can be separated into three main sub-circuits: the input sub-circuit that keeps traces of the incoming spikes is comparable to the basic circuit of figure 2, which itself is similar to those of relevant previous works [10–15]. The energy requirement per STDP event of this sub-circuit can be estimated by the cost of producing the traces. They dissipate $\sim C_p V_{\text{spike}}^2 \sim 10 \text{ nF } (1 \text{ V})^2 \sim 10 \text{ nJ}$, where we assumed a typical order of magnitude value for the spike amplitude V_{spike} . As expected, from the circuit similarity this estimate is consistent with those that can be obtained from the previous works [10–15]. The second sub-circuit is the one denoted as ADC in figure 5. This is comparable to the memristive integrate-and-fire neuron circuit [8], whose energy requirement per STDP event can be estimated as the cost of producing the train of pulses. Thus, we have $\sim CV_p^2 n_p \sim 5 \text{ nF } (5 \text{ V})^2 10 \sim 1250 \text{ nJ}$, where

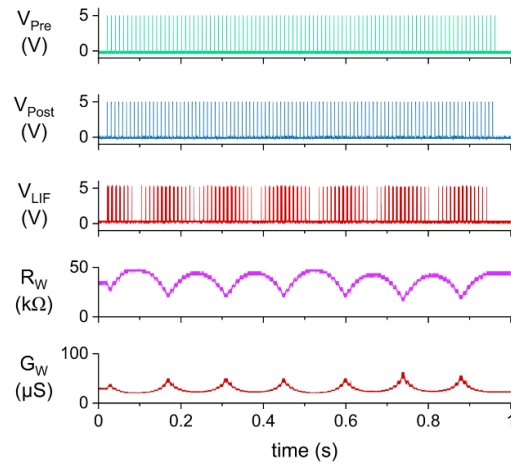


Figure 8. Periodic modulation of the synaptic conductance (Causal STDP). Two top panels: spiking behavior of neuron green and neuron blue. They are excited with currents of similar value that produces a small difference in their frequency. This leads to ‘beating’ where spikes of blue and green periodically advance and lags each other. Then the delay is short, there are many pulses generated by the active dendrite compartment (ADC). This produces the periodic increase and decrease of the resistance R_W (digipot), hence, the changes in the synaptic conductance G_W .

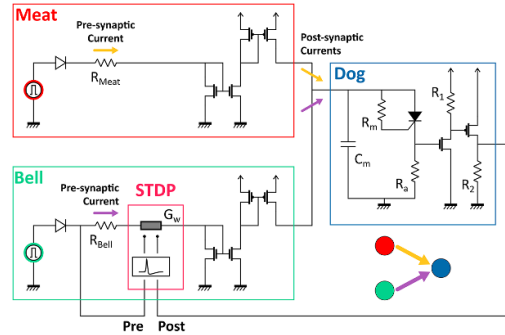


Figure 9. Pavlov Dog Circuit. The STDP block represents the sub-circuit of figure 5 with the conductance G_W (digipot) explicitly shown. The component parameter values are: $R_{\text{meat}} = 25 \text{ k}\Omega$, $R_{\text{bell}} = 20 \text{ k}\Omega$, $R_m = 10 \text{ k}\Omega$, $R_a = 5 \text{ k}\Omega$, $C_m = 20 \text{ nF}$, $R_1 = 100 \text{ k}\Omega$, $R_2 = 1 \text{ k}\Omega$, the transistors of the current mirrors and pulse shaper are ZNVL110A (NMOS) and ZVP331A (PMOS).

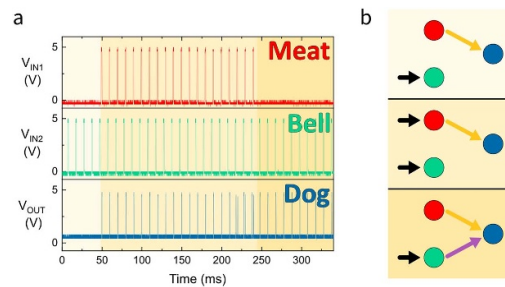


Figure 10. Classic conditioning (associative learning) implemented with the synaptic circuit of figures 5 and 9. Panel (a) shows spiking traces. The neuron-synapse configuration is shown in panel (b). Initially, neuron green (IN2) that represents the ‘bell’ is excited but not able to drive a ‘dog’s salivation’ response in the output neuron blue (OUT), i.e. they do not have an effective synaptic coupling. At $t = 50 \text{ ms}$ an unconditioned stimulus neuron red (IN1) that represents ‘meat’ is activated and drives the output due to their intrinsic strong synaptic connection (orange arrow). The spikes of the green neuron become causally paired to those of the blue neuron (i.e green spikes briefly precedes the blue ones). This causal correlation provokes the increase of the synaptic efficacy between neuron green (pre) and neuron blue (post), which by the STDP mechanism increase their synaptic coupling (purple arrow). After $t = 250 \text{ ms}$ of training we observe that the neuron green (bell) is now able to provoke the response of neuron blue (dog’s salivation).

V_p denotes the amplitude of the pulses and we assumed the number of pulses in the train $n_p = 10$. For concreteness we have adopted the values of the parameters in figures 5 and 6 above. This choice, done mostly for the sake of illustration, is likely to be overestimating the energy requirement by an order of magnitude. In fact, the value of V_p could be easily reduced to 3 V and $n_p = 10$ is more an upper bound

than a typical expected value. Nevertheless, our estimate for the second sub-circuit is significantly larger than that of the first one. This indicates that the ADC sub-circuit is likely to dominate the energy cost in an actual implementation. The third sub-circuit corresponds to the digipot. Its energy cost in a typical STDP update can be estimated from its datasheet and the duration of the update from figure 6. We estimate for the energy, $\sim 50 \mu\text{A} \cdot 5 \text{ V} \cdot 5 \text{ ms} \sim 1250 \text{ nJ}$, which is of a similar order of magnitude as that of the previous sub-circuit.

The apparent high energy cost of the active dendrite sub-circuit can be justified by the improvement that it provides in the shape of the STDP function $\Delta G_W(\Delta\tau)$. In fact, comparing the results from the basic STDP circuit in figure 2(b) to those of figure 7, we observe that the latter captures much better the decay of ΔG_W with $\Delta\tau$ and in the biological case [2]. The digipot sub-circuit also has a relative high energy cost. In this case, this can be justified on the merit of simplicity. Hence, this solution may be useful for the implementation of plastic spiking neural networks with a relatively small number of neurons, with that number depending on the specific context of the application (e.g brain-machine-interface, control systems, etc). A digipot based solution may not be appropriate for large scale networks due to both the energy cost and the large footprint. In fact, this is one of the reasons for the current ongoing interest in the research of non-volatile memristors [16–19].

4. Conclusion

We have introduced a bio-inspired neuromorphic circuit that implements STDP. Similarly to most previous implementations our hardware model has input stages where the arrival of the pre- and post- spikes are detected and respective temporal traces are created. However, a novel distinctive feature of our hardware model is that the detected delay is transformed into a sequence of pulses by an IF stage, which we call the ADC. This ADC is bio-inspired as it encodes the delay in a train of pulses, which is sent ‘upstream’ to modify the synapse. The number of pulses is inversely related to the delay, realizing the timing-dependent modulation of the synaptic change, which is a characteristic of STDP that is rarely captured by hardware models.

Another original feature of our work is that we implement the synaptic with a digipot, which avoids shortcomings of conventional models using a capacitor and a subthreshold transistor. Moreover, Our digipot solution has the merits of wide availability, reliability and simple implementation. It therefore served well our present goal of introducing a novel STDP synaptic circuit model. Nevertheless, as mentioned before, if low-energy consumption and small device footprint is a concern, the digipot component may eventually be replaced by non-volatile memristors. One further merit of our circuit model is that such a replacement should be rather straightforward, once the rapidly evolving memristive technology becomes mature.

Our hardware model is based on conventional widely available and affordable components, which is practical for networks of up to few hundreds of synapses. This scale may not be sufficient for complex artificial intelligence applications. Nevertheless, since it is based on analog hardware, it is well adapted for applications such as brain-machine-interfaces where the ‘machine’ needs to run in real time and to incorporate learning plasticity.

While the network scale of our present implementation is relatively modest, there are no a priori impediments to port the circuit to VLSI CMOS. In fact, all our circuit components are rather standard for VLSI CMOS, including the current mirrors, the SR flip-flop (implemented with two NOR gates) and the digipot. The thyristor of the integrate-and-fire sub-circuit of the ADC has already been ported to VLSI CMOS in [21].

The main appeals of our STDP circuit is its simplicity and bio-inspiration, as it nicely captures the inverse relation between timing and synaptic change. Moreover, unlike VLSI implementations, our analog electronics approach does not suffer from long design delays nor requires complex fabrication procedures. It is fully modifiable and all its timescales can be easily tuned, as shown in the present work. Therefore, it may open a new way to a reliable and practical implementation of learning in intelligent spiking neural networks.

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Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: <https://github.com/Integrative-Neuroscience/ActiveDendrite-STDP> [22].

Author contributions

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Conceptualization (lead), Formal analysis (lead), Funding acquisition (lead), Investigation (lead), Methodology (equal), Project administration (lead), Supervision (lead), Writing – original draft (lead), Writing – review & editing (lead)

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